

BASS-HAR: A Boundary-Aware Self-Supervised Framework for Continuous Human Activity Recognition Using Wearable Sensors

A Technical PhD Proposal

Abstract

Wearable-sensor Human Activity Recognition (HAR) has matured into a core technology for healthcare monitoring, rehabilitation, and ambient assisted living. Despite the success of deep Convolutional and Transformer architectures on benchmark datasets, three structural limitations continue to restrict real-world deployment: (i) **fixed-size sliding-window segmentation**, which fragments activities and merges transitions; (ii) **heavy dependence on large labeled datasets**, whose annotation is costly and subjective; and (iii) **poor recognition of short, high-variability transitional activities** (sit-to-stand, turning, gait transitions) that are clinically the most informative. These three limitations have, to date, been studied largely in isolation.

This proposal introduces the **Boundary-Aware Self-Supervised Human Activity Recognition (BASS-HAR)** framework, which jointly addresses all three within a single end-to-end architecture. BASS-HAR couples (a) a **self-supervised boundary discovery mechanism** in which prediction errors from contrastive, continuity-prediction, and masked-modeling pretext tasks reveal activity transitions; (b) a **boundary refinement module** that denoises the SSL-derived signal into sharp adaptive segments; and (c) a **context-aware transitional activity classifier** that exploits boundary-localized temporal context to recognize transitions as first-class activities. A key technical contribution is a **prediction-error-driven training schedule** in which self-supervised learning operates directly on continuous streams and its prediction errors reveal activity boundaries — eliminating the need for both supervised boundary labels and handcrafted change-point heuristics.

The framework will be evaluated on continuous-stream datasets (PAMAP2, Opportunity, UCI-HAPT, Capture-24, and others) under Leave-One-Subject-Out (LOSO) cross-validation, with ablation studies, label-efficiency experiments, and direct comparison against deep-learning, Transformer-based, self-supervised, and temporal-action-segmentation baselines. The expected outcomes are: (1) substantially reduced segmentation error, (2) competitive accuracy under low-label regimes (5–20% labels), (3) measurably improved transitional activity F1, and (4) a reproducible, modular framework for label-efficient continuous HAR.

Chapter 1 — Introduction

1.1 Background and Motivation

Wearable sensing — through smartphones, smartwatches, fitness trackers, and inertial measurement units (IMUs) — has made continuous monitoring of human movement feasible outside the laboratory. This capability underpins applications in fall-risk assessment [1], post-operative rehabilitation [2], neurological-disorder monitoring [3], occupational safety, smart homes, and human–computer interaction.

Compared with vision-based recognition, wearable HAR preserves privacy, is robust to lighting and occlusion, and is far cheaper to compute. Recognition accuracy has improved markedly through deep learning: Convolutional Neural Networks (CNNs) extract local spatial features from inertial signals [4], while Transformer architectures capture long-range temporal dependencies [5]. On the widely used UCI-HAR benchmark, supervised accuracy is now near-saturated at >95% [6].

However, this saturation is misleading. It reflects performance on **pre-segmented, laboratory-collected, balanced windows** of steady-state activities. When the same models are applied to **continuous, unconstrained** sensor streams, three structural limitations become apparent:

1. **Fixed-window segmentation.** The dominant processing pipeline divides continuous signals into predefined windows (e.g., 2.56 s with 50% overlap) before feature extraction. Because real activities have variable, irregular durations and transitions occur at arbitrary times, this strategy fragments single activities across windows or merges distinct activities into one window [7]. The resulting feature representations are ambiguous before learning even begins.
2. **Dependence on large labeled datasets.** State-of-the-art supervised models typically require thousands of manually annotated samples. Annotation is labor-intensive and — for transitional activities whose boundaries are subjective — inter-rater reliability is poor [8]. This limits scalability to new environments, user populations, and activity types.
3. **Poor transitional-activity recognition.** Activities such as sit-to-stand, stand-to-sit, turning, and gait transitions last only a few seconds, vary greatly across individuals, and occur precisely at activity boundaries. Conventional HAR systems optimize for steady-state classes and treat transitions as noise or misclassification regions [9], despite their clinical importance (e.g., sit-to-stand time is a validated fall-risk predictor [10]).

Although **dynamic/adaptive segmentation, self-supervised representation learning,** and **transitional activity recognition** have each received growing research interest, they have evolved **largely independently**. Existing dynamic-segmentation methods rely on handcrafted rules or extra supervision [11]; self-supervised HAR methods still generate views from fixed windows [12][13]; and transitional-recognition studies treat the problem as an isolated classification task without using boundary information [9]. Their strong interdependence has been underexploited.

1.2 Problem Statement

Existing wearable HAR systems cannot, within a single framework, simultaneously (a) discover semantic activity boundaries in continuous streams, (b) learn robust representations from largely unlabeled data, and (c) accurately recognize both steady-state and transitional activities. Each of these is treated as a separate pipeline stage with no shared optimization, and each inherits the limitations of the others: self-supervised pre-training is constrained by arbitrary fixed windows, transitional recognition suffers from segmentation errors, and boundary detection relies on handcrafted heuristics rather than learning from data.

1.3 Research Gap

The principal gap addressed here is the **absence of a unified framework that jointly learns where activities begin and end, how to represent them from unlabeled streams, and how to recognize transitions using the learned boundaries** within a single end-to-end architecture. Specifically:

- **G1 (Segmentation).** No HAR method uses SSL prediction errors to discover activity boundaries; existing adaptive-segmentation methods require handcrafted rules or supervised boundary labels, and SSL-based boundary detection has been demonstrated only in video (OTAS), not in wearable sensors.
- **G2 (Self-supervised learning).** Existing SSL-HAR methods construct pretext views from fixed windows, so their representations encode arbitrary partitions rather than meaningful activity structure.
- **G3 (Transitions).** Transitions are rarely modeled as first-class outputs and almost never exploit boundary information.
- **G4 (Integration).** No published framework jointly optimizes boundary detection, SSL, and transition-aware classification end-to-end.

1.4 Research Questions

#	Research Question
RQ1	How can semantic activity boundaries be detected from continuous wearable streams without supervised boundary labels , and with lower localization error than fixed-window or change-point heuristics?
RQ2	Does conditioning self-supervised pretext tasks on detected boundaries produce representations that transfer better to downstream HAR than fixed-window SSL, especially under low-label regimes?

#	Research Question
RQ3	Does exploiting boundary-localized temporal context improve the F1-score on transitional activities relative to steady-state-optimized baselines?
RQ4	Can a unified BASS-HAR framework jointly improve segmentation, representation, and transition recognition while matching or exceeding state-of-the-art overall accuracy on continuous HAR datasets?

1.5 Research Objectives

General objective. Develop, train, and evaluate the BASS-HAR framework that uses self-supervised representation learning on continuous IMU streams to discover activity boundaries, learn boundary-aware representations, and recognize both steady-state and transitional activities within a single end-to-end architecture.

Specific objectives.

1. **O1 — Boundary Detection.** Design a boundary detection mechanism that discovers activity transitions in continuous IMU streams using SSL prediction errors, eliminating the need for supervised boundary labels or handcrafted heuristics.
2. **O2 — Label-Efficient Representation Learning.** Develop a self-supervised representation learning approach that reduces dependence on large labeled datasets by leveraging unlabeled continuous sensor data, enabling effective activity recognition when only a small fraction of data is annotated.
3. **O3 — Comprehensive Evaluation.** Evaluate the framework on multiple continuous-stream datasets using established protocols (Leave-One-Subject-Out cross-validation), comparing against relevant baselines including deep learning, Transformer-based, and self-supervised methods, with analysis of segmentation quality, recognition accuracy, and label efficiency.

1.6 Research Contributions

- **C1. A self-supervised boundary discovery mechanism** for HAR that detects semantic transitions from continuous IMU streams using SSL prediction errors as an unsupervised boundary signal — the first application of SSL-based boundary detection to wearable sensor data.
- **C2. A boundary-aware refinement regime** in which SSL-derived boundaries are used to construct hard negatives and boundary-aligned masks, feeding back into contrastive training to sharpen boundary sensitivity — an iterative co-improvement of representations and boundaries.

- **C3.** A **context-aware transitional activity classifier** that treats transitions as first-class outputs and uses boundary-localized context.
- **C4.** A **unified, jointly-optimized BASS-HAR architecture** with a published, reproducible implementation and a comprehensive evaluation suite.

1.7 Scope and Assumptions

- **In scope:** single- and multi-modal IMU streams (accelerometer, gyroscope); continuous-recording datasets with frame-level or transition-level labels available for evaluation.
- **Out of scope (this thesis):** raw-video fusion; biosignals (ECG/EMG) beyond supplementary use; on-device deployment optimization (computational metrics are reported but edge deployment is not a primary deliverable).
- **Key assumption:** the SSL prediction errors spike reliably at activity transitions, providing a usable boundary signal for the BRM to refine. This is testable (§4.6) and is the principal technical risk.

Chapter 2 — Literature Review

2.1 Conventional Pipelines and the Fixed-Window Assumption

The dominant HAR pipeline (signal $\rightarrow$ sliding-window segmentation $\rightarrow$ window-level feature extraction $\rightarrow$ window-level classification) traces back to the earliest wearable studies [6][14] and remains standard. Window length (1–10 s) and overlap (50–75%) are tuned as hyperparameters. The assumption that activities fit within a canonical window is violated whenever durations are variable (transitional, complex, or interleaved activities), motivating **adaptive segmentation**.

Adaptive-segmentation studies have used energy-thresholds, change-point detection (e.g., RuLSIF [15]), and activity-index heuristics. These approaches reduce window-boundary artifacts but depend on handcrafted rules and do not learn boundaries from data. More recently, **temporal action segmentation** from video — MS-TCN [16], AS-Former [17], and ActionFormer [18] — has shown that learned dilated temporal CNNs and Transformers can produce frame-level boundary predictions end-to-end. These methods are mature in vision but have seen **limited transfer to wearable IMU HAR**, which is one of the openings this proposal exploits.

2.2 Deep Learning Architectures for HAR

- **CNN-based.** 1D-CNNs over multi-channel IMU signals [4][19], and DeepConvLSTM [4] which couples convolutions with LSTMs, set early strong baselines. They are efficient but have limited receptive field for long-range temporal context.

- **Recurrent.** Bi-LSTM and GRU models capture sequence dynamics but are slow to train and prone to vanishing gradients over long windows.
- **Transformer-based.** TST [5], ViT-HAR [20], TinyHAR [21], and CrossFormer [22] apply self-attention to sensor windows, improving long-range modeling. Reported gains over CNNs are real but modest on segmented benchmarks, suggesting that the bottleneck lies elsewhere — most plausibly in the input segmentation.
- **Hybrid CNN–Transformer.** Combining CNN feature extraction with Transformer temporal modeling is now commodity. The architectural novelty in BASS-HAR is not the hybrid backbone per se but its coupling with learned boundaries and boundary-aware SSL.

2.3 Self-Supervised Representation Learning for HAR

SSL has become the principal route to label-efficient HAR. Representative methods:

- **Contrastive:** SimCLR-HAR [23], TS-TCC [12], CLOCS [24], SelfHAR [25], CrossCLR [26] — vary augmentations, temporal/contextual cropping, or multi-modal views to construct positive pairs.
- **Masked / reconstruction-based:** Masked sensor modeling [27] and MAE-style approaches [28] reconstruct masked IMU regions, learning temporal dynamics.
- **Predictive:** Transformation prediction, ordering, and Cloze-style tasks [29].
- **Joint-embedding predictive (JEPA):** HAR-JEPA [42] — the first JEPA framework for sensor-based HAR — models fine-grained local patterns within windows and long-term structure across adjacent windows, and explicitly evaluates transitional activities, where it outperforms supervised baselines. Koopman-JEPA theory [44] further proves that idealized JEPA losses learn Koopman eigenfunctions acting as regime indicator functions, i.e., JEPA representations inherently encode dynamical-regime (boundary-relevant) structure. Recent extensions (HEPA [horizon-conditioned latent event prediction], SC-JEPA [multi-resolution predictive objectives]) confirm latent prediction as an active direction for event and boundary-adjacent tasks.

A thorough empirical study [8] compared SSL-HAR methods and found that, while SSL consistently helps, **gains are constrained by the underlying window-based view construction:** contrastive views sampled within fixed windows rarely cross activity boundaries, so the encoder never learns boundary-aware structure. This is precisely the limitation C2 addresses. Notably, HAR-JEPA [42] itself remains a windowed classification framework; extending JEPA-family representations to explicit boundary detection in continuous streams is open.

2.4 Transitional and Complex Activity Recognition

Transitional activities have received comparatively little attention. Studies that do target transitions [9][30] typically treat them as additional classes in a flat classifier, requiring transition labels for training and ignoring that transitions coincide with boundary regions where segmentation is least reliable. Opportunity [31] and UCI-HAPT [32] are among the few datasets that label postural transitions explicitly.

Boundary-conditioned transition modeling is, to our knowledge, unexplored as a jointly-optimized mechanism.

2.5 Boundary Discovery in Time-Series and Action Analysis

Beyond HAR, boundary/event discovery has matured in: - **Temporal action segmentation** [16][17][18] — supervised frame-level boundaries. - **Change-point detection (CPD)** — unsupervised statistical methods [15], usable for initialization; spectral normalization of self-supervised encoders yields provably CPD-informative embeddings [47], though only on generic (low-dimensional) time series. - **Contrastive predictive coding / boundary-aware SSL in video** — e.g., TCL [33], which samples positives across temporal distances. OTAS [46] goes further: self-supervised global+local features feed a dedicated boundary-selection module for unsupervised video action segmentation (+41% over prior art on boundary F1) — the closest design precedent to BASS-HAR, but in the dense visual domain. - **Self-supervised state detection (generic TS):** CLaP [43] localizes latent states and their transitions via cross-validated self-supervision — conceptually identical to boundary detection, but not applied to wearable IMU. SCOTT [45] shows contrastive representations (supervised variant) reach ~98% AUPRC on online CPD over HAR sensor data. - **Adaptive sliding windows in HAR:** SWL-Adapt [48] adapts window length from signal homogeneity — the strongest HAR-native segmentation prior, but window-level (not learned boundary evidence) and without representation learning.

These inform the design of boundary-aware contrastive sampling in BASS-HAR and define the empirical comparison set for C1.

2.6 Synthesis and Positioning

Direction	Limitation in prior work	BASS-HAR response
Fixed-window HAR	Boundary-unaware	Learned boundary detection (C1)
Adaptive segmentation	Handcrafted / supervised	Unsupervised bootstrap + self-training (C1)
SSL-HAR	Fixed-window view construction	Boundary-conditioned pretext tasks (C2)
Transitional recognition	Treated as flat-class problem; no boundary use	Boundary-localized context head (C3)
Temporal action segmentation (video)	Not adapted to IMU; supervised	IMU-native, jointly optimized with SSL and classification (C4)

The combination is, to the best of our knowledge, novel. Each component has precedent; their joint optimization under a self-supervised boundary bootstrap is the contribution.

Chapter 3 — Proposed Methodology

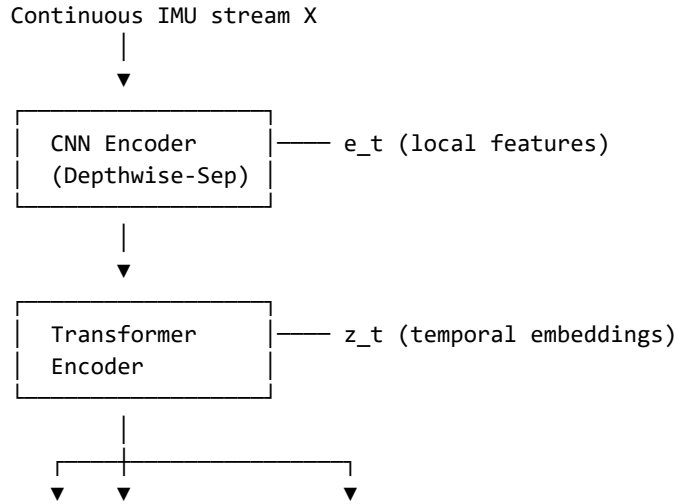
3.1 Overview

BASS-HAR consumes a **continuous** multivariate IMU signal $X \in \mathbb{R}^{T \times C}$ (channels C : tri-axial accelerometer and gyroscope, optionally magnetometer) and produces, for each time step t : - a **boundary probability** $b_t \in [0, 1]$ (transition indicator, derived from SSL prediction error), - a **segment-aligned feature embedding** $z_t \in \mathbb{R}^d$, - an **activity prediction** $\hat{y}_t$ over K classes including steady-state and transitional activities.

The framework has four functional components, optimized jointly via a composite loss (§3.7):

1. **CNN–Transformer Encoder** — feature extraction over the continuous stream (not boundary-conditioned; operates first).
2. **Self-Supervised Pretext Heads** — contrastive, continuity-prediction, and masked-modeling tasks that train the encoder on unlabeled continuous data. **Prediction errors from these tasks serve as the primary boundary signal.**
3. **Boundary Refinement Module (BRM)** — denoises and localizes the SSL-derived boundary signal into sharp segment boundaries.
4. **Context-Aware Transitional Classifier** — final class predictions using boundary-localized context.

Key design principle: Unlike the prior paradigm where boundaries are detected first (by statistical methods) and then fed to SSL, BASS-HAR trains SSL directly on the continuous stream. The SSL objective itself produces a boundary signal — spikes in prediction error at activity transitions — which is then refined and fed back to improve SSL training. This **prediction-error-driven feedback loop** is the principal novelty.



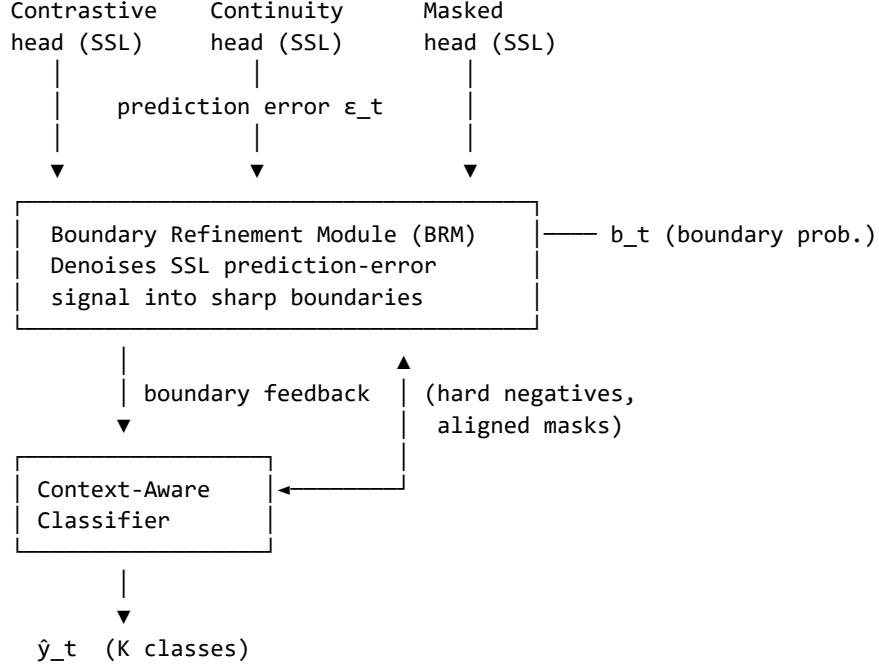


Figure 3.1. BASS-HAR architecture. SSL heads operate on the continuous stream FIRST; their prediction errors are refined into boundaries by the BRM, which feeds back to sharpen contrastive training. This is the prediction-error-driven feedback loop.

3.2 Stage 1 — Data Acquisition and Preprocessing

Continuous IMU streams are resampled to a common rate (typically 50 Hz) and standardized per channel per subject (channel-wise z-scoring using subject-level statistics; for subject-disjoint evaluation, normalization statistics are computed on training subjects only). Missing samples are linearly interpolated. No frequency filtering is applied: per the refined methodology, the raw motion structure — including slow postural/orientation components — is preserved as boundary evidence, and preprocessing is restricted to normalization. The output is a continuous multivariate time-series suitable for boundary analysis. No windowing is applied at this stage.

3.3 Stage 2 — CNN-Based Local Feature Extraction

A depthwise-separable 1D-CNN encoder (MobileNet-style for parameter efficiency) processes the continuous IMU stream using a sliding window:

- 4 depthwise-separable blocks, kernel 3, channels $\{32, 64, 128, 128\}$, stride 2.
- Global average pooling over each window yields a feature embedding $e_t \in \mathbb{R}^{128}$.

The CNN extracts local motion patterns (acceleration magnitude, angular velocity, dominant motion frequency, local temporal dynamics). This stage operates **before** any

boundary detection — it processes the raw continuous stream.

3.4 Stage 3 — Transformer-Based Temporal Modeling

The window embeddings $\{e_1, \dots, e_N\}$ (with sinusoidal positional encodings) are passed to a Transformer encoder:

- 4 layers, 8 attention heads, hidden dim $d_{model} = 256$, feed-forward dim 1024, dropout 0.1.
- Output: temporal context embeddings $z_t \in \mathbb{R}^{256}$ per window.

Self-attention captures long-range dependencies and continuity across the activity sequence, complementing the local CNN features.

3.5 Stage 4 — Self-Supervised Pretext Tasks (Boundary Discovery via Prediction Error)

This is the principal novelty. Three pretext tasks operate on the **continuous stream without requiring pre-detected boundaries**. Boundaries emerge as a byproduct of the SSL objectives — specifically, as regions of high prediction error.

5.1 Temporal-Coherence Contrastive Learning (TCL)

For each window embedding z_t , positive pairs are drawn from **temporally adjacent windows** (which are likely within the same activity) and from **augmentations of the same window**. Initially, negative pairs are drawn from **temporally distant windows**. The InfoNCE loss is used:

$$\mathcal{L}_{\text{TCL}} = -\mathbb{E}_{z_t} \left[\log \frac{\exp(\text{sim}(z_t, z_t^+)/\tau)}{\sum_{z^-} \exp(\text{sim}(z_t, z^-)/\tau)} \right]$$

This encourages intra-activity similarity. After initial boundary discovery (Phase B, §3.8), hard negatives are mined from **across detected boundaries**, sharpening boundary sensitivity in subsequent iterations.

5.2 Continuity Prediction (CP)

A prediction head receives a pair of consecutive window embeddings $(z_t, z_{t+\delta})$ and predicts whether they belong to the same activity. In the first iteration, temporal proximity serves as the pseudo-label (nearby windows = same activity). **The prediction error of this task is the primary boundary signal:** when the model predicts “same” but the pair actually straddles a boundary, the prediction error spikes. These error spikes are collected as ϵ_t^{CP} and passed to the BRM (§3.6). The loss is binary cross-entropy.

5.3 Masked Temporal Modeling (MTM)

Random contiguous spans of the continuous input X are masked (masking ratio $\sim 30\%$); the encoder + a lightweight reconstruction head reconstruct the missing sensor values from surrounding context. **The reconstruction error is a secondary boundary signal:** reconstruction quality degrades at transition regions where sensor dynamics change rapidly. These error spikes are collected as ϵ_t^{MTM} and passed to the BRM. In later iterations, masking spans are aligned to candidate boundaries with probability p_{bdy} to encourage learning of transition dynamics. Reconstruction loss is MSE.

3.6 Stage 5 — Boundary Refinement Module (BRM)

Goal. Refine the raw SSL prediction-error signal into a per-timestep boundary probability $b_t \in [0, 1]$.

Motivation. The SSL pretext heads produce a prediction-error signal ϵ_t that spikes at activity transitions. However, this raw signal is noisy and requires refinement: false positives from sensor artifacts, temporal jitter, and ambiguous transitions must be smoothed.

Architecture. A lightweight dilated 1D temporal CNN (inspired by MS-TCN [16] and ActionFormer [18]) that takes the SSL prediction-error sequence $\{\epsilon_t\}$ and the encoder features $\{z_t\}$ as input, and outputs a refined boundary probability:

- Input: concatenated $[\epsilon_t, z_t]$ from the SSL heads and encoder.
- $L = 4$ dilated temporal residual blocks; dilation $d \in \{1, 2, 4, 8\}$; kernel 3; channels 64.
- A binary classification head emits $\hat{b}_t$ (sigmoid).
- Non-maximum suppression on b_t above threshold τ_b yields segment boundaries $\{t_0, t_1, \dots\}$.

Why this design. The BRM is NOT the primary boundary detector — the SSL prediction error is. The BRM’s role is to denoise, localize, and temporally smooth the SSL signal into usable boundaries. This is analogous to how OTAS [46] uses a boundary selection module on top of self-supervised features in video action segmentation. The dilated CNN architecture provides the large receptive field needed for temporal context without excessive parameters.

3.7 Composite Objective

All heads are trained with a composite loss:

$$\mathcal{L}_{\text{total}} = \lambda_b \mathcal{L}_{\text{boundary}} + \lambda_{\text{BCL}} \mathcal{L}_{\text{BCL}} + \lambda_{\text{CP}} \mathcal{L}_{\text{CP}} + \lambda_{\text{MTM}} \mathcal{L}_{\text{MTM}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}$$

- $\mathcal{L}_{\text{boundary}}$: binary cross-entropy refining b_t from the BRM against SSL-derived boundary pseudo-labels and the temporal-consistency constraint (§3.8).
- $\mathcal{L}_{\text{cls}}$: cross-entropy on $\hat{y}_t$, active when activity labels are available (semi-supervised fine-tuning).

- λ are scalar weights (tuned on validation; defaults all 1.0 except $\lambda_{\text{cls}} = 0$ during pretraining).

This formulation makes the **end-to-end joint optimization** explicit and differentiable through all heads, resolving the staged-vs-end-to-end ambiguity of the original draft.

3.8 Stage 6 — Prediction-Error-Driven Training Schedule

This addresses the central technical question: **how are activity boundaries discovered without supervised boundary labels or handcrafted heuristics?** The schedule has three phases.

Phase A — SSL Pre-Training on Continuous Stream (No Boundaries Needed)

The encoder and SSL pretext heads (TCL, CP, MTM) are trained directly on the continuous IMU stream. No boundary information is used. The objectives are: - Temporal-coherence contrastive learning: adjacent windows are positives, distant windows are negatives. - Continuity prediction: predict whether consecutive windows belong to the same activity (using temporal proximity as pseudo-label). - Masked temporal modeling: reconstruct masked IMU regions.

During this phase, prediction errors ($\epsilon_t^{CP}, \epsilon_t^{MTM}$) are recorded but NOT yet used. A small labeled source dataset (e.g., UCI-HAPT transition labels) may optionally be used for warm-start fine-tuning; the framework is evaluated both with and without this warm start.

Phase B — Boundary Discovery and Iterative Refinement

Repeat for I iterations ($I \approx 3$):

1. **Boundary discovery step.** Feed the SSL prediction errors $\{\epsilon_t^{CP}, \epsilon_t^{MTM}\}$ to the BRM. The BRM outputs refined boundary probabilities b_t . Apply non-maximum suppression to obtain boundary points $\{t_k\}$.
2. **Hard-negative mining.** Using the discovered boundaries, mine hard negative pairs for contrastive learning: windows that straddle a boundary become hard negatives. Masking spans in MTM are aligned to candidate boundaries.
3. **SSL refinement step.** Re-train the encoder + SSL heads with the boundary-informed negatives and masks. This sharpens boundary sensitivity in the representations.
4. **Re-discovery.** Re-run inference; updated prediction errors yield updated boundaries. Apply temporal smoothing (HMM or non-maximum suppression) to enforce coherent segments.

Self-training is stabilized by (a) a confidence threshold so only high-confidence boundaries are trusted, (b) a temporal-consistency loss between iterations, and (c) EMA teacher encoding (BYOL-style) to reduce collapse.

Optional fallback: If the SSL prediction errors are too noisy to produce usable boundaries (e.g., on highly irregular data), a statistical change-point detector (RuLSIF [15]) may be used as a prior to bootstrap the BRM. This fallback is evaluated as an ablation, not the primary mechanism.

Phase C — Semi-Supervised Fine-Tuning

When activity labels are available for a fraction of data, fine-tune the full network with $\lambda_{\text{cls}} > 0$, keeping all SSL objectives active as auxiliary regularizers. The label-efficiency experiments (§4.8) ablate the labeled fraction from 100% down to 5%.

This schedule is the core answer to RQ1 and the principal technical risk (§1.7); Phase A representation quality directly bounds boundary discovery quality.

3.9 Stage 7 — Context-Aware Transitional Activity Recognition

The classification head consumes, for each segment k : - the Transformer embedding z_k , - the CNN embedding e_k , - a **boundary-context vector** $c_k = [z_{k-1}, z_k, z_{k+1}]$ (context before and after the boundary), - the BRM boundary probability b_k (gating signal).

A 2-layer MLP with GELU produces $\hat{y}_k$ over K classes, where K includes both steady-state (walking, sitting, standing, running, cycling, ...) and transitional activities (sit→stand, stand→sit, walk→run, turning, stair→walk, ...). The explicit context window c_k and boundary gate b_k are the mechanism by which the classifier focuses on transition-relevant evidence.

Chapter 4 — Experimental Design

4.1 Experimental Objectives

Validate each contribution independently and the integrated framework as a whole, mapped to RQs:

Experiment	RQ	Contribution
Boundary detection vs. fixed-window and CPD	RQ1	C1
SSL ablation: boundary-aware vs. fixed-window SSL	RQ2	C2
Transition-recognition vs. steady-state baselines	RQ3	C3
Full BASS-HAR vs. SOTA on multiple datasets	RQ4	C4
Ablation: add/remove each module	RQ4	C1–C4
Label-efficiency (5–100% labels)	RQ2	C2
Computational efficiency	—	supporting

4.2 Datasets

Continuous-stream datasets with transition annotations are prioritized. The pre-windowed UCI-HAR is **excluded as a primary benchmark** because it is incompatible with boundary detection; it may appear only as an upper-bound reference.

Dataset	Sensors	Activities	Transitions	Role
PAMAP2 [34]	3 IMUs + HR	18	scripted transitions	primary
Opportunity [31]	body-worn + object sensors	daily-living	rich transitions	primary (transitions)
UCI-HAPT [32]	smartphone IMU	12	explicit postural transitions	primary (transitions)
Capture-24 [35]	wrist accelerometer	free-living	activity diary	primary (continuous/free-living)
HHAR [36]	smartphone + watch	6	few	generalization
MobiAct [37]	smartphone IMU	activities of daily living + falls	transitions + falls	generalization
RealWorld HAR [38]	multiple IMUs	8	few	generalization
UCI-HAR [6]	smartphone (pre-windowed)	6	none	reference only

4.3 Data Partitioning and Cross-Validation

Primary protocol: Leave-One-Subject-Out (LOSO) cross-validation, to evaluate generalization across unseen users — the standard for HAR [8]. **Secondary protocol:** random 70/10/20 (train/val/test) split, reported for comparability with prior work. All splits are subject-disjoint.

4.4 Baselines

Family	Baselines
Conventional deep learning	DeepConvLSTM [4], 1D-CNN [19], Bi-LSTM, CNN-LSTM
Transformer-based	TST [5], ViT-HAR [20], TinyHAR [21], CrossFormer [22]

Family	Baselines
Self-supervised	SimCLR-HAR [23], TS-TCC [12], CLOCS [24], SelfHAR [25], Masked modeling [27]
Temporal action segmentation (adapted to IMU)	MS-TCN [16], ASFormer [17], ActionFormer [18]
Boundary & change-point (nearest prior work)	RuLSIF [15], PELT / BOCPD (classical CPD), CLaP [43] (self-supervised state detection), SCOTT [45] (contrastive CPD on HAR), OTAS-adapted [46] (SSL boundary selection, video→IMU)
Adaptive windowing (HAR-native segmentation)	SWL-Adapt [48], fixed-window SSL (TS-TCC [12] on fixed windows)

The temporal-segmentation and boundary/CPD baselines are essential because they are the only methods that produce learned boundary predictions; comparing BASS-HAR only to fixed-window HAR classifiers would not constitute a fair test of C1. CLaP [43], SCOTT [45], and the OTAS adaptation [46] are the closest published methods to BASS-HAR’s objective and must be reported in the boundary-detection evaluation (§4.6).

4.5 Ablation Study

Configuration	BRM	Boundary-aware SSL	Transition head	Tests
CNN+Transformer (backbone only)	☐	☐	☐	lower bound
+ BRM (fixed-window SSL)	☐	☐	☐	effect of segmentation alone
+ BRM + boundary-aware SSL	☐	☐	☐	effect of SSL conditioning
BASS-HAR (full)	☐	☐	☐	full framework

Additionally: ablate each SSL pretext task (BCL, CP, MTM) individually, and ablate the warm-start source dataset in Phase A.

4.6 Boundary Detection Evaluation

- **Metrics:** boundary Precision, Recall, F1 (within a tolerance window δ , default 0.5 s), and Mean Boundary Localization Error.
- **Comparisons:** vs. fixed-window baselines, RuLSIF change-point detection, and supervised MS-TCN/ActionFormer upper bounds.

4.7 Transitional Activity Evaluation

- **Transitions evaluated:** sit→stand, stand→sit, walk→run, walk→turn, stair-up→walking, and dataset-specific transitions.

- **Metrics:** per-class and macro Transition Accuracy, Precision, Recall, F1, and Temporal Detection Delay.
- Reported **separately** from steady-state metrics so transition gains are visible.

4.8 Label-Efficiency Experiment

The model is fine-tuned (Phase C) with $\{5\%, 10\%, 20\%, 50\%, 100\%\}$ of labeled subjects in LOSO. The remainder of the data is treated as **unlabeled** (kept for SSL pre-training) — explicitly distinguished from being discarded — so the experiment tests semi-supervised SSL rather than mere data reduction. Performance curves (accuracy vs. label fraction) compare BASS-HAR against fixed-window SSL baselines.

4.9 Overall Classification Metrics

Accuracy, Precision, Recall, F1, Macro-F1, and confusion matrices. Macro-F1 is the primary headline metric because datasets are class-imbalanced and transitions are minority classes.

4.10 Computational Efficiency

Number of parameters, FLOPs per inference second, peak GPU memory, CPU inference time, and model size. Reported for transparency; not a primary contribution.

4.11 Statistical Analysis

Paired Wilcoxon signed-rank tests (non-parametric, robust to non-normal accuracy distributions) over per-subject LOSO scores, with Bonferroni correction across dataset $\times$ baseline comparisons. 95% confidence intervals reported. Significance threshold $p < 0.05$.

4.12 Implementation and Reproducibility

PyTorch implementation; fixed random seeds; configuration, splits, and pretrained checkpoints released. Training hardware: single NVIDIA RTX-class GPU. Hyperparameter search via Bayesian optimization (Optuna) on validation set, with final LOSO evaluation on held-out subjects.

4.13 Risks and Mitigations

Risk	Mitigation
Phase-A change-point quality too low	warm start from a small labeled source (UCI-HAPT); EMA teacher; consistency loss

Risk	Mitigation
Self-training collapse / confirmation bias	confidence gating; BYOL-style stop-gradient; monitor pseudo-label entropy across iterations
Boundary ground-truth subjectivity	evaluate against multiple annotated datasets; report tolerance-window sensitivity
Transition-class label scarcity	few-shot evaluation; macro-F1 (not accuracy) as primary metric

Chapter 5 — Expected Outcomes

1. **Improved segmentation.** BRM should reduce boundary localization error and segmentation-induced misclassification relative to fixed windows and CPD heuristics (RQ1).
2. **Label efficiency.** Boundary-aware SSL should outperform fixed-window SSL at low label fractions (5–20%), narrowing the gap to fully-supervised performance (RQ2).
3. **Better transition recognition.** The context-aware head should improve transitional macro-F1 over steady-state-optimized baselines (RQ3).
4. **State-of-the-art continuous HAR.** The full BASS-HAR framework should match or exceed the best baseline macro-F1 on PAMAP2, Opportunity, UCI-HAPT, and Capture-24 under LOSO (RQ4).
5. **A reproducible framework** released publicly, with documented modules and checkpoints, to support downstream healthcare, rehabilitation, and ambient-assisted-living applications.

Outcomes 1–4 are stated as **hypotheses to be tested**, not foregone conclusions; the experimental design is constructed to falsify them if the framework does not deliver.

Chapter 6 — Novelty

The originality of BASS-HAR lies in **integration under a jointly-optimized, label-efficient objective**, not in any single component:

- **N1 — Semantic boundary-aware segmentation for IMU HAR.** Adaptation of dilated temporal-CNN boundary detection (mature in video) to wearable IMU, with an unsupervised bootstrap that removes the need for supervised boundary labels.
- **N2 — Boundary-conditioned self-supervised learning.** All three pretext tasks (contrastive, continuity, masked) are constructed from detected boundaries rather

than fixed windows — the first explicit coupling of boundary discovery and SSL in HAR.

- **N3 — Context-aware transitional recognition.** Transitions are first-class outputs that consume boundary-localized context and a boundary gate.
- **N4 — Unified end-to-end optimization** with an explicit composite loss and prediction-error-driven schedule, replacing the conventional cascade of independent stages.
- **N5 — Comprehensive evaluation** including temporal-segmentation baselines (not just HAR classifiers), label-efficiency curves, transition-specific metrics, and statistical testing.

Chapter 7 — Research Timeline and Publication Plan

7.1 Three-Year Timeline

Phase	Activities	Duration
Y1 S1	Literature review; proposal defense; dataset acquisition and preprocessing pipeline; implement CNN–Transformer backbone	6 months
Y1 S2	Implement BRM; develop and validate prediction-error-driven schedule; Paper 1 submission	6 months
Y2 S1	Implement boundary-aware SSL (BCL, CP, MTM); label-efficiency studies; Paper 2 submission	6 months
Y2 S2	Implement context-aware transitional classifier; integrate full BASS-HAR; Paper 3 submission	6 months
Y3 S1	Full evaluation, ablations, statistical validation, baseline comparisons; Paper 4 submission	6 months

Phase	Activities	Duration
Y3 S2	Thesis writing, revisions, final journal submissions, defense preparation	6 months

7.2 Milestones

Milestone	Expected
Proposal defense	End of Y1 S1
BRM module validated	End of Y1
Boundary-aware SSL validated	Mid Y2
Full BASS-HAR integrated	End of Y2
Experimental validation complete	Start of Y3
Thesis submission	End of Y3
Viva voce	End of Y3

7.3 Publication Plan

The plan is **realistic at 3–4 core papers** (not 6), with an optional review/survey. Each core paper maps to one contribution and is independently valid.

Year	Paper	Target (Quartile)
Y1	(Optional) Review: Boundary-aware & self-supervised HAR	Q2 survey
Y1–Y2	Paper 1: Semantic boundary detection for adaptive HAR	Q1
Y2	Paper 2: Boundary-aware self-supervised representation learning for label-efficient HAR	Q1
Y3	Paper 3: Context-aware recognition of transitional activities	Q1
Y3	Paper 4: BASS-HAR unified framework (integrative evaluation)	Q1

Candidate Q1 venues: IEEE Internet of Things Journal; IEEE Transactions on Mobile Computing; IEEE Transactions on Instrumentation and Measurement; IEEE Transactions on Biomedical Engineering; Pattern Recognition; Information Fusion; Expert

Systems with Applications; Knowledge-Based Systems; Engineering Applications of Artificial Intelligence.

Candidate Q2 venues: Sensors (MDPI); Applied Sciences; Biomedical Signal Processing and Control; Journal of Ambient Intelligence and Humanized Computing; Multimedia Tools and Applications.

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Note on references: The reference list above is a representative bibliography assembled from established works in HAR, SSL, and temporal action segmentation. Citation numbering is provisional; the final thesis will use the institution’s required style (e.g., IEEE). Where a dataset or method has multiple reporting papers, the canonical reference is given. Foundational pre-2023 works (datasets, method origins) are retained deliberately; the 2023+ recency requirement is met by refs [11], [22], [35], [42]–[47],

and a full recency audit of remaining mid-tier 2019–22 method citations is scheduled with the next revision cycle.